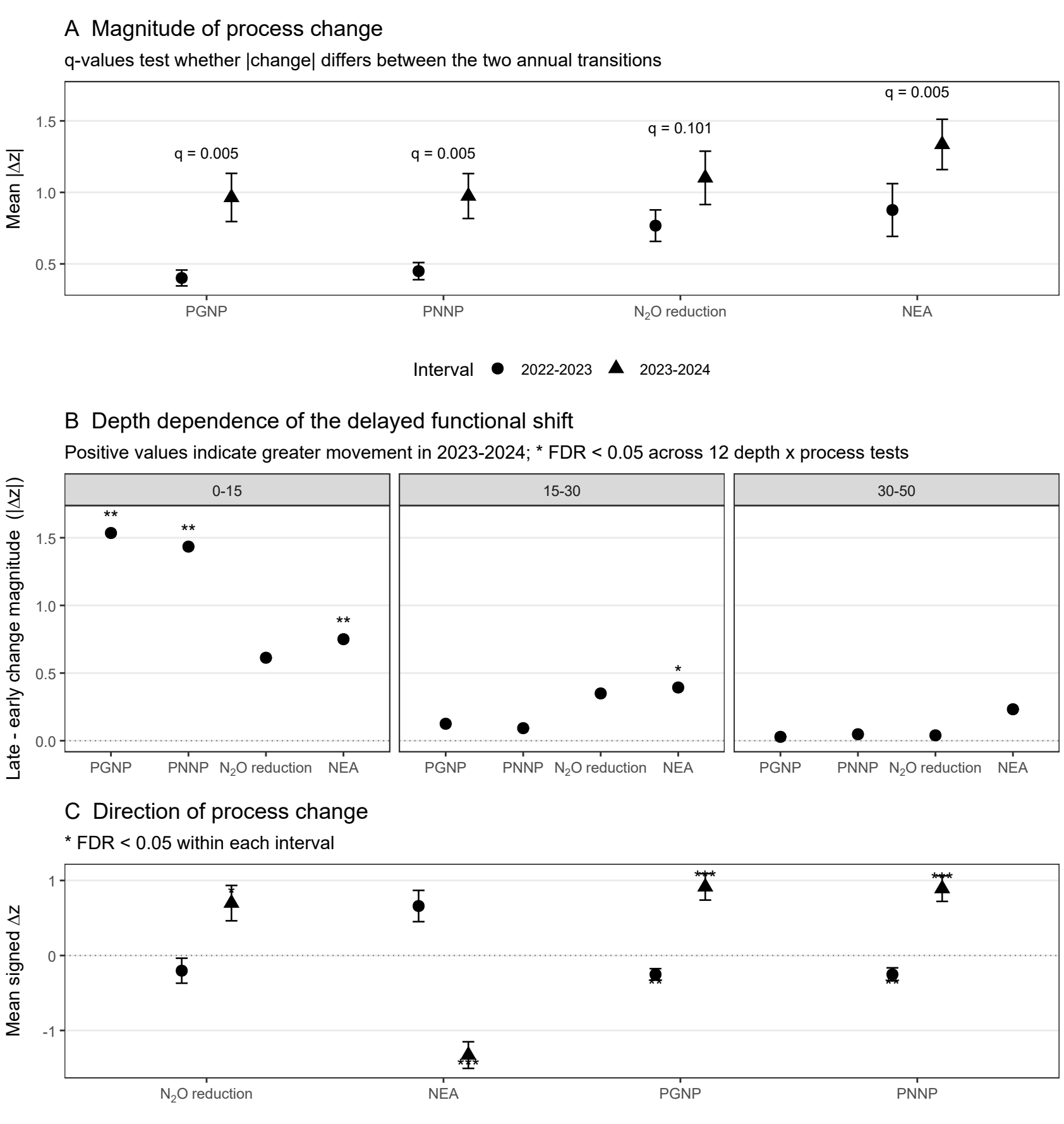
A Magnitude of process change
q-values test whether |change| differs between the two annual transitions
Mean |Δz|
1.5
1.0
0.5
q = 0.005
q = 0.005
q = 0.101
q = 0.005
PGNP
PNNP
N2O reduction
NEA
Interval
2022-2023
2023-2024
B Depth dependence of the delayed functional shift
Positive values indicate greater movement in 2023-2024; * FDR < 0.05 across 12 depth x process tests
Late - early change magnitude (|Δz|)
0-15
15-30
30-50
1.5
1.0
0.5
0.0
**
**
**
*
PGNP
PNNP
N2O reduction
NEA
C Direction of process change
* FDR < 0.05 within each interval
Mean signed Δz
1
0
-1
*
***
**
***
**
***
N2O reduction
NEA
PGNP
PNNP